Important

Outline to be used for content formation for Data Science, machine learning, Research based(CSor DS) or Artificial Intelligence based FYPs.

Use the template given for CS students but use these contents for making your chapters. CS students who are building applications must follow the chapters as mentioned in the template.

Note: **<u>This is not the template for writing but just a guideline how student doing projects in the above-mentioned fields must have these chapters in their thesis.</u>**

Chapter 1 Introduction

This chapter describes the background of the problem, and the motivation behind selecting the research area carried out in this thesis. It should also provide an overview of the thesis structure.

Background

Problem Statement

Research Questions

Thesis Objectives

Contributions

Thesis Organization

Chapter 2 Theoretical Background

The chapter should provide theoretical background required to understand the problem and the work.

Introduction

Subsection 2.2

Subsection 2.3

Subsection 2.4

Chapter 3

Literature Review

A detailed and comprehensive literature review. It can be divided into sections as well. For instance, one section could be about the solutions provided in the literature using classical machine, another section could be about the solutions using deep learning approaches, another could be about the use of natural language processing techniques and so on. It is not necessary that all the theses will have these sections in this chapter, it depends on the nature of the problem as well. Please change accordingly.

Literature review summary

A table summarizeing authors their works and limitations with proper citation

Gap analysis

Comparing the existing work/technologies and identifying the gap , your proposed work is going to fill. This section is very important for the students who are carrying research based FYP

Chapter 4 Methodology

This chapter describes the proposed framework/technique/model and experimental settings in detail.

Overview

Describe an overview of the proposed framework/model.

Proposed Framework/Model/Technique

Describe in detail the proposed framework/model/technique. Provide sufficient details about the work. The model architecture and other details can be provided here.

Methodology

Describe in detail how the experiments are performed. In the first para, provide an overview of the process/methodology.

Dataset

Describe how the dataset is collected or is it available online. Provide information about the data, its properties, features etc.

Data Pre-Processing

Feature Engineering

Feature Selection

Model Training

Evaluation Criteria

What evaluation criteria is used? Provide explanation and formulae.

Benchmark Algorithms

What are other benchmark algorithms against which you are comparing your proposed technique.

Chapter 5 Results and Discussion

Results

Present the results based on the evaluation criterion. Compare the performance of the proposed framework/model/technique with benchmark algorithms/techniques considering the various performance evaluation criteria.

Use table/charts to present your results.

Discussion

Please discuss the results. Give rationale for the results, why some are better than others, what can be the possible reasons. Why are some results not as expected? Justify the results, if possible. Provide context from the literature.

Chapter 6 Conclusion and Future Work

Conclusion

Future Work

References